Problem 1: Running Sum of 1d Array (easy)

Problem Statement

Given a one-dimensional array of integers, create a new array that represents the running sum of the original array.

The running sum at position `i` in the new array is calculated as the sum of all the numbers in the original array from the 0th index up to the `i`-th index (inclusive). Formally, the resulting array should be computed as follows: `result[i] = sum(nums[0] + nums[1] + ... + nums[i])` for each `i` from 0 to the length of the array minus one.

Examples

Example 1

- **Input**: `[2, 3, 5, 1, 6]`
- **Expected Output**: `[2, 5, 10, 11, 17]`
- **Justification**:
  - For i=0: 2
  - For i=1: 2 + 3 = 5
  - For i=2: 2 + 3 + 5 = 10
  - For i=3: 2 + 3 + 5 + 1 = 11
  - For i=4: 2 + 3 + 5 + 1 + 6 = 17

Example 2

- **Input**: `[1, 1, 1, 1, 1]`
- **Expected Output**: `[1, 2, 3, 4, 5]`
- **Justification**: Each element is simply the sum of all preceding elements plus the current element.

Example 3

- **Input**: `[-1, 2, -3, 4, -5]`
- **Expected Output**: `[-1, 1, -2, 2, -3]`
- **Justification**: Negative numbers are also summed up in the same manner as positive ones.

Try it yourself

Try solving this question here:

Java

```java
class Solution {
    public int[] runningSum(int[] nums) {
        int[] result = new int[nums.length];
        // TODO: Write your code here
        return result;
    }
}
```

Test Case | Run Code Result | Stdout

[3,1,4,2,2]

Use Example Testcases | Run | Submit

Please check the next lesson for a complete solution.